

THE TRIVIALITY CONJECTURE FOR SPHERICAL COCHAINS

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ABSTRACT. We prove the triviality conjecture of Heuts and Land: for $n \geq 1$ and an $\mathbb{E}_{n+2}$ -ring spectrum A , the augmented cochain algebra $C^*(S^n; A)$ is $\mathbb{E}_{n+1}$ -trivial over A if and only if A is rational. The proof constructs, at each prime p , a D_p -structure on the original multiplication of $A_{(p)}$. A comparison between configuration spaces and representation spheres turns the operadic obstruction into a functional on cochains. Equivariant normalization and one additional little-disks direction give the required product and unit compatibilities. Hahn's chromatic-acyclicity theorem and the Hopkins–Smith nilpotence theorem then imply rationality.

1. INTRODUCTION

Let $n \geq 1$ and let A be an $\mathbb{E}_{n+2}$ -ring spectrum. The A -valued cochains on S^n , augmented by evaluation at the basepoint, have underlying A -module

$$C^*(S^n; A) \simeq A \oplus \Sigma^{-n} A.$$

The triviality question asks when this splitting identifies the cochain algebra with a square-zero extension. Following Heuts and Land [5, §§2.3, 4.1], we say that $C^*(S^n; A)$ is $\mathbb{E}_{n+1}$ -trivial over A if it is equivalent to the trivial square-zero extension in

$$\mathrm{Alg}_{\mathbb{E}_{n+1}}^{\mathrm{aug}}(\mathrm{Mod}_A).$$

The $\mathbb{E}_{n+2}$ -structure on A gives Mod_A its $\mathbb{E}_{n+1}$ -monoidal structure.

Heuts and Land proved that spherical cochains are $\mathbb{E}_n$ -trivial [5, Theorem A]. They conjectured that $\mathbb{E}_{n+1}$ -triviality characterizes rational coefficients [5, §4.1]. We prove this conjecture.

Theorem 1.1. *Let $n \geq 1$ and let A be an $\mathbb{E}_{n+2}$ -ring spectrum. Then*

$$C^*(S^n; A) \simeq A \oplus \Sigma^{-n} A \quad \text{in } \mathrm{Alg}_{\mathbb{E}_{n+1}}^{\mathrm{aug}}(\mathrm{Mod}_A),$$

with the trivial square-zero structure on the right, if and only if the unit-induced map $A \longrightarrow A \otimes H\mathbb{Q}$ is an equivalence.

The rational implication is [5, Lemma 4.10]. Heuts and Land also established the conjecture for $n \leq 2$, for bounded-below coefficients, and for coefficients whose given structure extends to an $\mathbb{E}_\infty$ -structure [5, Theorem 4.11]. Their Theorem 4.12 supplies the initial chromatic vanishing used here: triviality implies that both $H\mathbb{F}_p \otimes A$ and $KU/p \otimes A$ vanish for every prime p .

1.1. The construction of an extended multiplication. Fix a prime p and put $R = A_{(p)}$, with unit η_R . The additional input is Proposition 4.2: spherical triviality gives a D_p -structure on the original multiplication of R , in the sense of Hahn [4, Definition 1.2]. Hahn's Theorem 1.3 then propagates the height-one vanishing to every positive finite height. The Hopkins–Smith nilpotence theorem, applied to the unit of $R \otimes \mathrm{End}(\mathbb{S}/p)$, gives $R/p = 0$.

The construction of the D_p -structure has three stages. The primary arity- p operation for sphere-valued cochains, suspended n times, has target the sphere spectrum. Triviality over R makes its

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composite with η_R nullhomotopic. In Lemma 3.2, we identify the cofiber of this sphere-valued map with homotopy orbits of the dual of the unit sphere in n copies of the reduced permutation representation of Σ_p . This identification preserves the map from the sphere spectrum, so the nullhomotopy supplies a functional whose value on that map is η_R .

Next choose the least odd integer $m \geq n$. The centered map from $\text{Conf}_p(\mathbb{R}^m)$ to the corresponding representation sphere induces a p -local stable equivalence on Borel constructions (Proposition 2.1). Restricting along the inclusion of representation spheres and using this comparison produces a functional on the homotopy orbits of sphere-valued configuration-space cochains. Lemma 4.3 turns its unit condition into an equivariant left R -module retraction of the constant-function map.

Finally, compose this retraction with the adjoint of the arity- p little-disks action. The inclusion $\mathbb{E}_m(p) \rightarrow \mathbb{E}_{m+1}(p)$ is nonequivariantly nullhomotopic. It follows that the composite agrees with the original p -fold product after forgetting the group action. Substitution of the nullary unit gives the required equivariant unit identity. Restriction from Σ_p to C_p therefore produces the two compatibilities in Hahn's definition. The inequality $m + 1 \leq n + 2$ supplies the little-disks structure used in this last stage.

1.2. Conventions and notation.

Convention 1.2. In Sections 2–5, until the globalization in the proof of Theorem 1.1, fix a prime p and work in $\text{Sp}_{(p)}$. Write $\mathbb{S} = \mathbb{S}_{(p)}$ and use $\otimes$ for the smash product. A finite spectrum means a retract of a finite cell $\mathbb{S}_{(p)}$ -module. An action of a finite group G means an object of $\text{Fun}(BG, \text{Sp}_{(p)})$. All ring spectra and all little-disks algebras are unital.

For the configuration-space and operadic constructions, set

$$\begin{aligned} G &= \Sigma_p, & V &= \{(t_1, \dots, t_p) \in \mathbb{R}^p : \sum_i t_i = 0\}, \\ X_k &= \text{Conf}_p(\mathbb{R}^k), & Y_k &= S(kV). \end{aligned} \tag{1.1}$$

Here Conf_p denotes ordered configurations, $S(W)$ is the unit sphere of a real representation W , and $\mathbb{S}^W$ is its one-point compactification, stabilized and p -localized. The notation G is also used as a variable finite group in the general lemmas on equivariant duality and normalization.

For a G -space Z of finite homotopy type, write

$$DZ_+ = F(\Sigma_+^\infty Z, \mathbb{S}), \quad U(Z) = (DZ_+)_{hG}.$$

The constant-function map $c_Z : \mathbb{S} \rightarrow DZ_+$ is equivariant, with trivial action on its source. Let $a : \mathbb{S} \rightarrow \mathbb{S}_{hG}$ be induced by the basepoint of BG , and define

$$\gamma_Z : \mathbb{S} \xrightarrow{a} \mathbb{S}_{hG} \xrightarrow{(c_Z)_{hG}} U(Z). \tag{1.2}$$

If $j : Z \rightarrow Z'$ is equivariant, contravariant functoriality gives

$$U(j)\gamma_{Z'} \simeq \gamma_Z. \tag{1.3}$$

We use η_R for the unit of a ring spectrum R , and c_R for the constant-function map $R \rightarrow C^*(Z; R)$ when Z is specified. The little-disks space $\mathbb{E}_k(p)$ is equivariantly homotopy equivalent to X_k ; an explicit comparison is given in Lemma 4.4.

2. CONFIGURATION SPACES AND REPRESENTATION SPHERES

Centering and normalizing a configuration defines a G -equivariant map

$$j_k : X_k \longrightarrow Y_k, \quad (x_i)_{i=1}^p \longmapsto \frac{(x_i - \bar{x})_{i=1}^p}{(\sum_{i=1}^p \|x_i - \bar{x}\|^2)^{1/2}}, \quad \bar{x} = \frac{1}{p} \sum_{i=1}^p x_i. \tag{2.1}$$

The denominator is nonzero because the points are distinct.

Proposition 2.1. *If k is odd, the map j_k induces equivalences*

$$\Sigma_+^\infty(X_{k,hG})_{(p)} \xrightarrow{\simeq} \Sigma_+^\infty(Y_{k,hG})_{(p)}, \quad (2.2)$$

$$U(Y_k) \xrightarrow{U(j_k)} U(X_k). \quad (2.3)$$

For $p = 2$, both assertions hold for every $k \geq 1$.

For odd k and odd p , the cohomology of each Borel construction is generated by pullbacks from BG . We will compute these groups and use the fact that j_k lies over BG to prove that it induces an isomorphism. Finite p -local models then give (2.2), and the norm identifies its dual with (2.3).

2.1. Finite models and norm maps.

Lemma 2.2. *For every $k \geq 1$, the spaces X_k and Y_k have finite G -CW homotopy models. The action on X_k is free. Every isotropy subgroup of Y_k has order prime to p .*

Proof. The action on ordered configurations is free. If the stabilizer of a point of Y_k had order divisible by p , it would contain an element of order p . Such an element of Σ_p is a p -cycle, whose fixed subspace in V , and hence in kV , is zero. It therefore fixes no unit vector.

Let $\mathcal{D} \subset Y_k$ be the union of the collision subspaces $\{x_i = x_j\} \cap Y_k$. Centering and normalizing gives an equivariant homeomorphism

$$X_k \cong \mathbb{R}^k \times \mathbb{R}_{>0} \times (Y_k \setminus \mathcal{D}),$$

with trivial action on the first two factors. Choose a finite G -invariant central hyperplane arrangement in kV containing defining hyperplanes for the collision subspaces, with total intersection zero. The resulting spherical polyhedral decomposition, followed by barycentric subdivision, gives a finite equivariant triangulation K of Y_k . The collision locus is a subcomplex L , and the action has no inversions on simplices.

Inside $\text{sd } K$, take the full subcomplex spanned by vertices corresponding to simplices of K not contained in L . The complement $|K| \setminus |L|$ deformation retracts onto this subcomplex: discard the barycentric weights at the other vertices and renormalize the remaining weights. Their sum is positive off $|L|$, and linear interpolation gives the deformation. This construction is equivariant, and the resulting finite subcomplex is free because it lies in $Y_k \setminus \mathcal{D}$. These triangulations give the required G -CW models. $\square$

Lemma 2.3. *Let Z have a finite G -CW model whose isotropy groups have order prime to p . Then $\Sigma_+^\infty(Z_{hG})_{(p)}$ is a finite spectrum, and the norm gives an equivalence*

$$(DZ_+)_{hG} \xrightarrow{\simeq} (DZ_+)^{hG} \simeq F(\Sigma_+^\infty(Z_{hG}), \mathbb{S}). \quad (2.4)$$

This identification is natural in equivariant maps between such spaces.

Proof. The homotopy orbits of an orbit cell $G/H_+ \otimes \mathbb{S}^d$ are $\Sigma_+^\infty BH \otimes \mathbb{S}^d$. If $|H|$ is prime to p , transfer annihilates the positive-degree $\mathbb{Z}_{(p)}$ -homology of BH by the unit $|H|$. Thus the augmentation $\Sigma_+^\infty BH_{(p)} \rightarrow \mathbb{S}$ is an equivalence: its cofiber is connective and $H\mathbb{Z}_{(p)}$ -acyclic, so stable Hurewicz at its first possible nonzero homotopy group makes the cofiber zero. Finite cellular induction proves the finiteness of $\Sigma_+^\infty(Z_{hG})_{(p)}$.

For the norm, recall that the Tate functor is exact and vanishes on induced spectra. Under the induction–coinduction identifications, the norm of an induced spectrum is the identity [7, Lemma I.3.8(i) and its proof]. The trivial p -local H -sphere is a retract of $\bigoplus_{h \in H} \mathbb{S}$ in $\text{Fun}(BH, \text{Sp}_{(p)})$: the diagonal followed by the fold has underlying map $|H|$, hence is an equivalence, and its equivariant inverse gives the retraction. Inducing to G shows that every orbit cell with prime-to- p isotropy lies in the thick subcategory generated by induced G -spectra.

The finite permutation spectrum $\Sigma_+^\infty(G/H)$ is equivariantly self-dual. Since duality is exact on finite spectra and reverses suspensions, DZ_+ lies in the same thick subcategory. Its Tate spectrum therefore vanishes, so the norm is an equivalence. Finally, the colimit–limit adjunction gives the natural identification

$$F(\Sigma_+^\infty Z, \mathbb{S})^{hG} \simeq F((\Sigma_+^\infty Z)_{hG}, \mathbb{S}).$$

□

Lemma 2.4. *A map between finite p -local spectra which induces an isomorphism on $\mathbb{F}_p$ -homology is an equivalence.*

Proof. Let C be the cofiber. The universal coefficient sequence gives $H_i(C; \mathbb{Z}_{(p)})/p = 0$ for every i . These modules are finitely generated, so Nakayama’s lemma gives $H_i(C; \mathbb{Z}_{(p)}) = 0$. The spectrum C is bounded below; stable Hurewicz at its first possible nonzero homotopy group now gives $C = 0$. □

2.2. Cohomology over the symmetric group.

Lemma 2.5. *Let p be odd and $k = 2r + 1 \geq 3$. Write ι for the degree-zero, weight-one generator of the free $\mathbb{E}_k$ -algebra on a point. Then*

$$H_*(X_{k,hG}; \mathbb{F}_p) = \mathbb{F}_p\{\iota^p, Q^s \iota, \beta Q^s \iota : 1 \leq s \leq r\}, \quad (2.5)$$

with degrees 0 , $2s(p-1)$, and $2s(p-1) - 1$, respectively. The projection to BG is injective on homology and sends these classes, up to nonzero scalars, to the corresponding stable classes.

Proof. Cohen’s free-algebra homology theorem, in the form of [3, Theorem 16.4, §§16.1–16.3], applies to the generator ι . Its weight- p summand is the homology of $X_{k,hG}$. An operation of length two has weight at least p^2 . The Browder bracket of ι with itself has antisymmetry sign $(-1)^{1+(k-1)} = -1$, so it vanishes over $\mathbb{F}_p$. Consequently all Lie words of length at least two in this generator vanish. The weight- p part of the free-algebra basis consists of ι^p and the single operations in (2.5).

The endpoint $s = r$ consists of the top operations ξ and ζ , written as Q^r and βQ^r in [3, §16.2.1, p. 179]. Under inclusion into the free $\mathbb{E}_\infty$ -algebra, these map to the corresponding stable operations, up to the nonzero scalars determined by the conventions for the top operations; see [2, Chapter III, Theorem 1.3(1), p. 217]. The stable weight- p basis is ι^p together with $Q^s \iota, \beta Q^s \iota$ for all $s \geq 1$, so the displayed images are nonzero and independent.

The map to the stable weight- p space is the classifying map $X_{k,hG} \rightarrow B\Sigma_p$. Its injectivity on homology also follows directly from [2, Chapter III, Theorem 5.2, p. 246]: for odd k , pullback along this map is surjective on cohomology, with kernel the classes of degree greater than $(k-1)(p-1)$. □

Lemma 2.6. *Let p be odd and $k = 2r + 1 \geq 1$. There are classes u, v on BG , with $\mathbb{F}_p$ coefficients, of degrees*

$$|u| = 2(p-1), \quad |v| = 2p-3,$$

such that the projection $Y_{k,hG} \rightarrow BG$ induces a ring isomorphism

$$H^*(Y_{k,hG}; \mathbb{F}_p) \cong \frac{\mathbb{F}_p[u] \otimes \Lambda(v)}{(u^{r+1}, vu^r)}. \quad (2.6)$$

On the right, u and v denote the pullbacks of the classes on BG .

Proof. We compute the trivial- and sign-coefficient cohomology of BG , then the Euler map in the Gysin sequence of $Y_{k,hG} \rightarrow BG$.

Let $P = C_p \subseteq G$ be generated by a p -cycle. A periodic resolution gives

$$H^*(BP; \mathbb{F}_p) = \mathbb{F}_p[b] \otimes \Lambda(a), \quad |a| = 1, \quad |b| = 2, \quad b = \beta a.$$

Here b is the periodicity class, and $a^2 = 0$ by graded commutativity. The automorphism taking a generator to its λ th power multiplies both a and b by $\lambda \in \mathbb{F}_p^\times$.

With either trivial or sign coefficients, restriction to P is injective because the composite corestriction is multiplication by $[G : P] = (p-1)!$. Its image in positive degrees is the invariant subspace for $N_G(P)/P = \mathbb{F}_p^\times$. Indeed, in the double-coset formula for restriction, the intersections $P \cap gPg^{-1}$ are trivial unless $g \in N_G(P)$, so those terms vanish in positive degree. On an invariant class, the remaining terms sum to multiplication by $p-1$, which is invertible in $\mathbb{F}_p$. These restriction-transfer identities hold with local coefficients [1, Chapter III, §§9–10]. In degree zero, the invariant description follows directly from the coefficient module.

For trivial coefficients this gives

$$H^*(BG; \mathbb{F}_p) = \mathbb{F}_p[u] \otimes \Lambda(v), \quad \text{res}(u) = b^{p-1}, \quad \text{res}(v) = ab^{p-2}. \quad (2.7)$$

Put $s = (p-1)/2$. The sign of the permutation $x \mapsto \lambda x$ of $\mathbb{F}_p$ is λ^s , as follows by comparing the Vandermonde products before and after the permutation. Thus restriction with coefficients in $\mathbb{F}_p^{\text{sgn}}$ identifies the cohomology with the free $\mathbb{F}_p[u]$ -module on

$$b^s \quad \text{and} \quad ab^{s-1}. \quad (2.8)$$

The weight of a monomial under λ must be congruent to s modulo $p-1$ to cancel the coefficient action. Both generators have positive degree, including when $p=3$.

The determinant character of V is sgn . Its restriction to P is the sum of the oriented real two-planes underlying the complex characters of weights $1, \dots, s$. With a suitable orientation,

$$e(V)|_{BP} = s! b^s.$$

The coefficient is nonzero modulo p . Since k is odd, the orientation local system of kV is again sgn , and

$$e(kV)|_{BP} = (s!)^k b^{ks}.$$

The space $Y_{k,hG}$ is the sphere bundle of $EG \times_G kV$. The disk-sphere pair gives a Gysin sequence whose Euler maps are

$$H^{d-k(p-1)}(BG; \mathbb{F}_p^{\text{sgn}}) \xrightarrow{\cdot e(kV)} H^d(BG; \mathbb{F}_p).$$

The coefficient twist comes from the Thom isomorphism: over each cell of the base, the relative complex is shifted by the rank, and a change of trivialization acts by the orientation character. Pullback of the Thom class along the zero section is the Euler class, giving the displayed maps.

Restriction to BP is injective on source and target. There the Euler map is multiplication by a nonzero scalar times b^{ks} , which is injective on (2.8). Its image is generated by

$$b^{(k+1)s} = u^{r+1}, \quad ab^{(k+1)s-1} = vu^r,$$

up to nonzero scalars. The Gysin sequence therefore identifies $H^*(Y_{k,hG}; \mathbb{F}_p)$ with the quotient in (2.6). Pullback to the sphere bundle is a ring map, so this is an isomorphism of rings. $\square$

Proof of Proposition 2.1. For $p=2$, the difference vector, center, and distance give an equivariant homotopy equivalence $X_k \simeq S^{k-1} = Y_k$, with the transposition acting antipodally. After removing the contractible factors, this equivalence is j_k .

Suppose that p is odd. For $k=1$, the unordered configuration space is the convex region $x_1 < \dots < x_p$, hence is contractible. Lemma 2.6 with $r=0$ gives $H^*(Y_{1,hG}; \mathbb{F}_p) = \mathbb{F}_p$, so $j_{1,hG}$ is a mod- p homology equivalence.

For $k=2r+1 \geq 3$, Lemmas 2.5 and 2.6 give one-dimensional cohomology in the degrees

$$0, \quad 2s(p-1), \quad 2s(p-1)-1 \quad (1 \leq s \leq r),$$

and zero in every other degree. Both cohomologies are spanned by pullbacks from BG : on the configuration side, this follows by dualizing the injectivity in Lemma 2.5. Since the diagram

$$\begin{array}{ccc} X_{k,hG} & \xrightarrow{j_{k,hG}} & Y_{k,hG} \\ & \searrow & \swarrow \\ & BG & \end{array}$$

commutes, $j_{k,hG}^*$ is surjective. The dimensions agree, so it is an isomorphism.

Lemmas 2.2 and 2.3 show that the two p -local suspension spectra in (2.2) are finite. Lemma 2.4 therefore proves (2.2). Dualizing this equivalence and applying the natural norm identification (2.4) gives (2.3). $\square$

3. THE OPERADIC OBSTRUCTION AND ITS DISTINGUISHED MAP

For a p -local spectrum W and $1 \leq r \leq \infty$, let

$$D_p^{\mathbb{E}_r}(W) = (\Sigma_+^\infty \mathbb{E}_r(p) \otimes W^{\otimes p})_{hG}.$$

Here $\mathbb{E}_r(p)$ is the unbased little-disks space. This is the arity- p extended power for the nonunital stable operad of [5, §4.2]. For $k \geq 1$, define

$$\begin{aligned} M_k &= \Sigma^k D_p^{\mathbb{E}_{k+1}}(\mathbb{S}^{-k}), & N_k &= \Sigma^k D_p^{\mathbb{E}_\infty}(\mathbb{S}^{-k}), \\ C_k &= \text{cofib}(\mu_k : M_k \longrightarrow \mathbb{S}), \end{aligned} \tag{3.1}$$

where μ_k is the suspended arity- p structure map of the augmentation ideal of $C^*(S^k; \mathbb{S})$. Write $\gamma_k : \mathbb{S} \rightarrow C_k$ for the cofiber map.

Lemma 3.1. *If R is an $\mathbb{E}_{n+2}$ -ring spectrum and $C^*(S^n; R)$ is $\mathbb{E}_{n+1}$ -trivial over R , then there is a map $f : C_n \rightarrow R$ satisfying $f\gamma_n \simeq \eta_R$.*

Proof. The structure-map calculation of [5, §4.2, equation (2), p. 22], which applies to $\mathbb{E}_{n+2}$ -ring coefficients, identifies the relevant operation after base change with

$$M_n \xrightarrow{\mu_n} \mathbb{S} \xrightarrow{\eta_R} R.$$

By hypothesis, the nonunital augmentation ideal is a trivial $\mathbb{E}_{n+1}$ -algebra. Its arity- p operation is therefore nullhomotopic, so the displayed composite is nullhomotopic; see also the paragraph following the proof of [5, Proposition 4.18, p. 24]. A choice of nullhomotopy extends η_R across the cofiber of μ_n , giving the required map f . $\square$

Lemma 3.2. *For every $k \geq 1$ there is an equivalence*

$$\Phi_k : C_k \xrightarrow{\simeq} U(Y_k) \quad \text{such that} \quad \Phi_k \gamma_k \simeq \gamma_{Y_k}. \tag{3.2}$$

Proof. Let $\sigma^k : N_k \rightarrow D_p^{\mathbb{E}_\infty}(\mathbb{S}) = \mathbb{S}_{hG}$ be the suspension comparison, and put

$$T_k = \text{cofib}(\sigma^k : N_k \longrightarrow \mathbb{S}_{hG}).$$

We first identify T_k with $U(Y_k)$. A pushout square of Heuts–Land then reduces the comparison $C_k \rightarrow T_k$ to the case $k = 1$. The map of cofiber sequences defining this comparison will also identify the maps from $\mathbb{S}$.

The suspension cofiber. The collapse map $\Sigma_+^\infty \mathbb{E}_\infty(p) \rightarrow \mathbb{S}$ is an equivalence in the naive equivariant category. Since the permutation representation is the direct sum of V and the diagonal trivial representation, this gives

$$N_k \simeq (\mathbb{S}^{-kV})_{hG}.$$

Under this equivalence, σ^k is induced by the equivariant Euler map $\mathbb{S}^{-kV} \rightarrow \mathbb{S}$. Indeed, the suspension comparison is the strength map

$$S^k \otimes (W^{\otimes p})_{hG} \longrightarrow ((S^k \otimes W)^{\otimes p})_{hG},$$

formed using the reduced diagonal $S^k \rightarrow (S^k)^{\wedge p}$; this is the comparison used in [5, §3.1 and Lemma 4.14]. Regard S^k as the one-point compactification of $\mathbb{R}^k$. Its reduced diagonal is the compactification of the linear embedding

$$\mathbb{R}^k \hookrightarrow (\mathbb{R}^k)^p,$$

whose orthogonal complement is kV . Setting $W = \mathbb{S}^{-k}$ and canceling the diagonal trivial representation therefore identifies the comparison with the desuspension of $\mathbb{S} \rightarrow \mathbb{S}^{kV}$.

The equivariant Euler cofiber sequence and its dual are

$$\Sigma_+^\infty Y_k \longrightarrow \mathbb{S} \longrightarrow \mathbb{S}^{kV}, \quad \mathbb{S}^{-kV} \longrightarrow \mathbb{S} \xrightarrow{c_{Y_k}} DY_{k,+}.$$

Homotopy orbits preserve cofibers, hence

$$T_k \simeq U(Y_k), \tag{3.3}$$

with the map $\mathbb{S}_{hG} \rightarrow T_k$ corresponding to $(c_{Y_k})_{hG}$.

The cofiber comparison. Theorem 3.11 and Lemma 4.14 of [5] give a homotopy-coherent pushout square

$$\begin{array}{ccc} M_k & \longrightarrow & N_k \\ \beta^{k-1} \downarrow & & \downarrow \sigma^{k-1} \\ M_1 & \longrightarrow & N_1. \end{array} \tag{3.4}$$

The horizontal maps come from $\mathbb{E}_{k+1} \rightarrow \mathbb{E}_\infty$ and $\mathbb{E}_2 \rightarrow \mathbb{E}_\infty$, respectively. The vertical maps are the iterated wrong-way map β and suspension comparison σ .

The structure-map calculation in [5, pp. 22–24] identifies μ_k with the iterated wrong-way map β^k and gives a map from (3.4) to the square

$$\begin{array}{ccc} \mathbb{S} & \xrightarrow{a} & \mathbb{S}_{hG} \\ \parallel & & \parallel \\ \mathbb{S} & \xrightarrow{a} & \mathbb{S}_{hG}. \end{array} \tag{3.5}$$

Its four corner maps are μ_k , σ^k , μ_1 , and σ . To identify the horizontal map, use $\mathbb{E}_1(p) \simeq G$ and $D_p^{\mathbb{E}_1}(\mathbb{S}) \simeq \mathbb{S}$. The inclusion $\mathbb{E}_1(p) \rightarrow \mathbb{E}_\infty(p)$ then induces the map $\mathbb{S} \rightarrow \mathbb{S}_{hG}$ from a point of BG , namely the map a in (1.2). Thus the comparison is a map of the indicated squares with their specified horizontal arrows.

Both squares are pushouts. Taking pointwise cofibers gives a pushout square

$$\begin{array}{ccc} C_k & \longrightarrow & T_k \\ \downarrow & & \downarrow \\ C_1 & \longrightarrow & T_1. \end{array} \tag{3.6}$$

Since a pushout square in a stable category is also a pullback, it remains to show that its bottom arrow is an equivalence.

The case $k = 1$. Remarks 4.16–4.17 of [5] identify

$$(M_1 \xrightarrow{\mu_1} \mathbb{S}) \simeq (\Sigma^{-1}\mathbb{S}/p \xrightarrow{\beta} \mathbb{S}), \quad C_1 \simeq \mathbb{S} \simeq T_1,$$

where β is the integral Bockstein. We will show that the map $C_1 \rightarrow T_1$ induces an isomorphism on $H_0(-; \mathbb{F}_p)$.

For odd p , let $\iota_1 \in H_{-1}(\mathbb{S}^{-1}; \mathbb{F}_p)$ be the fundamental class. In the homology of the suspended extended powers, the computations of [5, Remarks 4.16–4.17] give the basis

$$Q^0(\iota_1), \quad \beta Q^0(\iota_1)$$

for $H_*(M_1; \mathbb{F}_p)$, in degrees 0 and -1 , respectively, and the basis

$$Q^s(\iota_1), \quad \beta Q^s(\iota_1), \quad s \geq 0,$$

for $H_*(N_1; \mathbb{F}_p)$. Here the notation includes the suspension of the Dyer–Lashof classes. The inclusion $M_1 \rightarrow N_1$ preserves these operations and their Bocksteins, so it sends the two displayed classes for M_1 to the classes of the same name in N_1 . In particular, it induces an isomorphism on H_{-1} . The maps μ_1 and σ send $Q^0(\iota_1)$ to the respective degree-zero generators, and hence induce isomorphisms on H_0 .

At $p = 2$, the stunted projective spectrum models in the proof of [5, Lemma 4.14] identify $M_1 \rightarrow N_1$ with the cellular inclusion $\mathbb{RP}_{-1}^0 \rightarrow \mathbb{RP}_{-1}^\infty$. The maps to $\mathbb{S}$ and $\mathbb{S}_{hG}$ are the corresponding quotient maps to $\mathbb{RP}_0^0$ and $\mathbb{RP}_0^\infty$. They give the same isomorphisms in degrees -1 and 0 .

Suppressing $\mathbb{F}_p$ from the notation, the long exact sequences of the two cofibers form a commutative diagram

$$\begin{array}{ccccccc} H_0(M_1) & \xrightarrow{\sim} & H_0(\mathbb{S}) & \longrightarrow & H_0(C_1) & \xrightarrow{\partial} & H_{-1}(M_1) \longrightarrow 0 \\ \downarrow & & \downarrow & & \downarrow & & \downarrow \sim \\ H_0(N_1) & \xrightarrow{\sim} & H_0(\mathbb{S}_{hG}) & \longrightarrow & H_0(T_1) & \xrightarrow{\partial} & H_{-1}(N_1) \longrightarrow 0. \end{array}$$

The zeros on the right follow from the connectivity of $\mathbb{S}$ and $\mathbb{S}_{hG}$. Exactness shows that both connecting maps are isomorphisms. The right vertical map is an isomorphism by the preceding computation, so $H_0(C_1) \rightarrow H_0(T_1)$ is an isomorphism as well. A map between p -local spheres with this property has degree a unit in $\mathbb{Z}_{(p)}$; thus $C_1 \rightarrow T_1$ is an equivalence.

It follows from (3.6) that $C_k \rightarrow T_k$ is an equivalence for every k . By its construction from a map of cofiber sequences, this equivalence carries γ_k to

$$\mathbb{S} \xrightarrow{a} \mathbb{S}_{hG} \longrightarrow T_k.$$

Under (3.3), this composite is γ_{Y_k} as defined in (1.2). Composing $C_k \rightarrow T_k$ with (3.3) gives the equivalence Φ_k and its required compatibility. $\square$

Corollary 3.3. *Let R satisfy the hypotheses of Lemma 3.1, and let*

$$m = \begin{cases} n, & n \text{ odd}, \\ n+1, & n \text{ even}. \end{cases} \quad (3.7)$$

There is a map $f_X : U(X_m) \rightarrow R$ such that

$$f_X \gamma_{X_m} \simeq \eta_R. \quad (3.8)$$

In addition, m is odd and $m+1 \leq n+2$.

Proof. Lemmas 3.1 and 3.2 give a map $f_n : U(Y_n) \rightarrow R$ with $f_n \gamma_{Y_n} \simeq \eta_R$. The equivariant inclusion $Y_n \hookrightarrow Y_m$ induces $U(Y_m) \rightarrow U(Y_n)$. Precomposing f_n with this map gives $f_m : U(Y_m) \rightarrow R$, and (1.3) implies

$$f_m \gamma_{Y_m} \simeq \eta_R.$$

Since m is odd, Proposition 2.1 makes $t = U(j_m) : U(Y_m) \rightarrow U(X_m)$ an equivalence. Define $f_X = f_m t^{-1}$. Applying (1.3) to j_m gives (3.8). The inequality $m+1 \leq n+2$ follows from (3.7). $\square$

4. A UNITAL EXTENSION OF THE MULTIPLICATION

Definition 4.1 (Hahn, [4, Definition 1.2]). A D_p -algebra is a spectrum R equipped with a homotopy-associative, homotopy-commutative unital multiplication and a map

$$\Psi_p : (R^{\otimes p})_{hC_p} \longrightarrow R$$

whose composite with $R^{\otimes p} \rightarrow (R^{\otimes p})_{hC_p}$ is homotopic to the p -fold product, and for which the diagram

$$\begin{array}{ccc} (\mathbb{S}^{\otimes p})_{hC_p} & \xrightarrow{(\eta_R^{\otimes p})_{hC_p}} & (R^{\otimes p})_{hC_p} \\ \varepsilon \downarrow & & \downarrow \Psi_p \\ \mathbb{S} & \xrightarrow{\eta_R} & R \end{array} \quad (4.1)$$

commutes up to homotopy. Here C_p acts on $R^{\otimes p}$ by cyclic permutation, and $\varepsilon : \Sigma_+^\infty BC_p \rightarrow \mathbb{S}$ is the augmentation.

Proposition 4.2. *Let R be a p -local $\mathbb{E}_{n+2}$ -ring spectrum, with $n \geq 1$. If $C^*(S^n; R)$ is $\mathbb{E}_{n+1}$ -trivial over R , then the original multiplication on R admits a D_p -structure.*

The functional of Corollary 3.3 will give an equivariant retraction of the constant-function map on configuration cochains. Composing this retraction with the little-disks action produces the required extension. One additional little-disks direction identifies its underlying map with the original product, while the nullary operadic composition supplies the equivariant unit identity.

4.1. Normalization of the functional.

Lemma 4.3. *Let R be a unital $\mathbb{E}_1$ -ring spectrum, let Z be a G -space whose underlying suspension spectrum is finite, and suppose that*

$$f : (DZ_+)_{hG} \longrightarrow R, \quad f\gamma_Z \simeq \eta_R.$$

There is an equivariant left R -module map

$$\bar{q} : C^*(Z; R) \longrightarrow R \quad \text{such that} \quad \bar{q}c_R \simeq \text{id}_R \text{ in } \text{Fun}(BG, \text{Mod}_R), \quad (4.2)$$

where c_R is the constant-function map and the target R has trivial G -action.

Proof. The colimit–constant-diagram adjunction identifies f with a map

$$\ell : DZ_+ \longrightarrow R \quad \text{in } \text{Fun}(BG, \text{Sp}_{(p)}).$$

By the definition of γ_Z , the underlying composite $\ell c_Z : \mathbb{S} \rightarrow R$ is homotopic to η_R . Finiteness gives a natural equivariant equivalence of left R -modules

$$C^*(Z; R) \simeq R \otimes DZ_+.$$

The free-module extension of ℓ is the equivariant left R -module map

$$q : R \otimes DZ_+ \xrightarrow{1 \otimes \ell} R \otimes R \xrightarrow{\mu} R.$$

Put $y = qc_R$, an endomorphism of R in $\text{Fun}(BG, \text{Mod}_R)$. Under

$$\text{Map}_{\text{Mod}_R}(R, R) \simeq \text{Map}_{\text{Sp}}(\mathbb{S}, R),$$

the underlying homotopy class of y corresponds to the unit. Thus its underlying R -module map is homotopic to id_R .

Equivalences in a functor category are detected objectwise, so y is an equivalence in $\text{Fun}(BG, \text{Mod}_R)$. Choose its inverse in that category and set $\bar{q} = y^{-1}q$. Then $\bar{q}c_R = y^{-1}y \simeq \text{id}_R$ equivariantly, as required. $\square$

4.2. The extra little-disks direction.

Lemma 4.4. *For $m \geq 1$, the standard inclusion $\mathbb{E}_m(p) \longrightarrow \mathbb{E}_{m+1}(p)$ is nonequivariantly nullhomotopic.*

Proof. The center maps from little-disks spaces to configuration spaces are equivariant homotopy equivalences compatible with inclusion of the ambient hyperplane. To see this explicitly, for centers z_i in the open unit ball put

$$\delta(z) = \frac{1}{4} \min(\{1 - \|z_i\|\}_i \cup \{\|z_i - z_j\| : i < j\}).$$

Choosing every radius equal to $\delta(z)$ gives a continuous equivariant section of the center map. For fixed centers the permissible radii form a convex set, defined by

$$r_i > 0, \quad r_i + r_j < \|z_i - z_j\|, \quad r_i < 1 - \|z_i\|.$$

Linear interpolation to the section therefore gives an equivariant deformation retraction. Radial homeomorphisms from open unit balls to Euclidean spaces can be chosen compatibly with hyperplane inclusions.

It remains to contract $\text{Conf}_p(\mathbb{R}^m) \rightarrow \text{Conf}_p(\mathbb{R}^{m+1})$. Define

$$H_t(x_1, \dots, x_p) = (((1-t)x_i, ti))_{i=1}^p, \quad 0 \leq t \leq 1. \quad (4.3)$$

At $t = 0$ this is the inclusion. For $t > 0$ the last coordinates are pairwise distinct, so H_t remains a configuration. At $t = 1$ the map is constant. $\square$

4.3. The extended product and its unit.

Proposition 4.5. *Let $m \geq 1$, and let R be an $\mathbb{E}_{m+1}$ -ring spectrum. Suppose there is an equivariant map as in (4.2), with $Z = \mathbb{E}_m(p)$ and $G = \Sigma_p$. Then the original multiplication on R admits a D_p -structure.*

Proof. The arity- p action of $\mathbb{E}_m$ on R is an equivariant map

$$\Sigma_+^\infty \mathbb{E}_m(p) \otimes R^{\otimes p} \longrightarrow R,$$

where G acts diagonally on the source and trivially on R . Its adjoint is an equivariant map

$$\nu : R^{\otimes p} \longrightarrow C^*(\mathbb{E}_m(p); R), \quad (4.4)$$

with the action on cochains given by precomposition. Define

$$\hat{\mu}_p = \bar{q}\nu : R^{\otimes p} \longrightarrow R \quad \text{in } \text{Fun}(BG, \text{Sp}_{(p)}).$$

We show that its underlying map is the p -fold product and that its restriction to the unit is equivariantly homotopic to η_R .

The underlying product. The $\mathbb{E}_m$ action is the restriction of the $\mathbb{E}_{m+1}$ action. After forgetting the group action, Lemma 4.4 therefore gives a homotopy

$$\nu \simeq c_R \mu_p, \quad (4.5)$$

where μ_p is evaluation of the $\mathbb{E}_{m+1}$ action at a fixed p -ary operation. Since $m+1 \geq 2$, the space $\mathbb{E}_{m+1}(p)$ is path connected. The chosen operation can thus be joined to an iterated binary operation in the standard $\mathbb{E}_1$ -suboperad. Hence μ_p is homotopic to the original p -fold product. Equation (4.2) now gives the nonequivariant homotopy

$$\hat{\mu}_p \simeq \bar{q} c_R \mu_p \simeq \mu_p.$$

The unit. The nullary operadic composition is the equivariant map

$$\mathbb{E}_m(p) \times \mathbb{E}_m(0)^p \longrightarrow \mathbb{E}_m(0) = *.$$

The unit compatibility of the $\mathbb{E}_m$ -algebra structure therefore yields

$$\nu\eta_R^{\otimes p} \simeq c_R\eta_R \quad \text{in } \text{Fun}(BG, \text{Sp}_{(p)}),$$

using the canonical equivariant identification $\mathbb{S}^{\otimes p} \simeq \mathbb{S}$. Composing with the equivariant retraction in (4.2) gives

$$\hat{\mu}_p\eta_R^{\otimes p} \simeq \eta_R \quad \text{in } \text{Fun}(BG, \text{Sp}_{(p)}). \quad (4.6)$$

Restrict $\hat{\mu}_p$ to $C_p \subseteq G$ and apply the homotopy-orbit adjunction to obtain $\Psi_p : (R^{\otimes p})_{hC_p} \rightarrow R$. The underlying-product calculation identifies its composite with $R^{\otimes p} \rightarrow (R^{\otimes p})_{hC_p}$ as the original product. Under the same adjunction, the map η_R between constant C_p -objects corresponds to $\eta_R\varepsilon$. Thus (4.6) gives the commutativity of (4.1). The $\mathbb{E}_{m+1}$ -structure, with $m+1 \geq 2$, already makes the original multiplication homotopy associative, homotopy commutative, and unital. These data define the required D_p -structure. $\square$

Remark 4.6. The left R -module structure is used to construct the retraction $\bar{q}$. The resulting D_p -structure is determined by the map Ψ_p together with its product and unit compatibilities in Definition 4.1.

Proof of Proposition 4.2. Choose m as in (3.7). Corollary 3.3 supplies a functional on $U(X_m)$ with the required value on γ_{X_m} . Lemma 4.3 therefore gives an equivariant left R -module retraction of the constant-function map on $C^*(X_m; R)$. The center-map equivalence $\mathbb{E}_m(p) \simeq X_m$ transports this retraction to $C^*(\mathbb{E}_m(p); R)$ and preserves constant functions. Since $m+1 \leq n+2$, the given structure on R restricts to an $\mathbb{E}_{m+1}$ -structure. Proposition 4.5 supplies the D_p -structure on its original multiplication. $\square$

5. CHROMATIC ACYCLICITY AND RATIONALITY

The D_p -structure of Proposition 4.2 allows us to propagate height-one acyclicity to every positive finite height. We then apply nilpotence to the unit of an endomorphism ring to prove that p is invertible.

Theorem 5.1 (Hahn [4, Theorem 1.3]). *If a spectrum R admits a D_p -algebra structure and $K(h, p) \otimes R = 0$ for some finite $h \geq 0$, then $K(h+1, p) \otimes R = 0$.*

Theorem 5.2 (Hopkins–Smith [6, Theorem 3(i)]). *Let B be a p -local ring spectrum. An element $x \in \pi_*B$ is nilpotent if its images in*

$$H_*(B; \mathbb{Q}), \quad H_*(B; \mathbb{F}_p), \quad K(h, p)_*B \quad (h \geq 1)$$

are all nilpotent.

The rational and mod- p theories in Theorem 5.2 are denoted $K(0, p)$ and $K(\infty, p)$ in [6]. The theorem applies, in particular, to unital associative ring spectra.

Lemma 5.3. *Let R be a p -local unital associative ring spectrum such that*

$$H\mathbb{F}_p \otimes R = 0, \quad K(h, p) \otimes R = 0 \quad \text{for every } h \geq 1.$$

Then $R/p = 0$; equivalently, multiplication by p on R is an equivalence.

Proof. Let $M = \mathbb{S}/p$ be the finite Moore spectrum. Composition equips $\text{End}(M) = F(M, M)$ with a unital associative multiplication, and hence

$$B = R \otimes \text{End}(M)$$

is a unital associative ring spectrum. Since M is dualizable, $\text{End}(M) \simeq DM \otimes M$. It follows from $H\mathbb{Q} \otimes M = 0$ that $H\mathbb{Q} \otimes B = 0$. The hypotheses on R give

$$H\mathbb{F}_p \otimes B = 0, \quad K(h, p) \otimes B = 0 \quad (h \geq 1).$$

Theorem 5.2 therefore makes $1_B \in \pi_0 B$ nilpotent. Since every positive power of 1_B is 1_B , the unit map of B is nullhomotopic. The unit identity

$$\mathrm{id}_B \simeq (B \simeq \mathbb{S} \otimes B \xrightarrow{\eta_B \otimes \mathrm{id}_B} B \otimes B \xrightarrow{\mu_B} B)$$

then gives $B = 0$.

Multiplication on R and evaluation on M make $R \otimes M$ a unital B -module. Its unit identity factors through $B \otimes (R \otimes M)$, so $B = 0$ implies $R \otimes M = 0$. Thus $R/p = 0$. $\square$

Proposition 5.4 (Reduction at a prime). *Let $n \geq 1$, let R be a p -local $\mathbb{E}_{n+2}$ -ring spectrum, and let m be the least odd integer at least n . Then $m + 1 \leq n + 2$, and the following implications hold:*

$$\text{spherical triviality} \implies f_X \gamma_{X_m} \simeq \eta_R,$$

$$(f_X \gamma_{X_m} \simeq \eta_R) + \mathbb{E}_{m+1}\text{-structure} \implies D_p\text{-structure on the original product,}$$

$$D_p\text{-structure} + H\mathbb{F}_p\text{-acyclicity} + K(1, p)\text{-acyclicity} \implies R/p = 0.$$

Together with the initial vanishings of [5, Theorem 4.12], these implications reduce the only-if direction of Theorem 1.1 to the passage from $A_{(p)}/p = 0$ for every p to the rationality of A .

Proof. The first implication is Corollary 3.3. For the second, apply Lemma 4.3 and Proposition 4.5. For the third, Theorem 5.1 propagates $K(1, p)$ -acyclicity to all positive finite heights, and Lemma 5.3 then gives $R/p = 0$. $\square$

Proof of Theorem 1.1. Assume that $C^*(S^n; A)$ is $\mathbb{E}_{n+1}$ -trivial over A . We apply Proposition 5.4 to prove that $A_{(p)}/p = 0$ at every prime, and then pass from these local vanishings to the rationality of A .

Vanishing of $A_{(p)}/p$. Fix a prime p and put $R = A_{(p)}$. Smashing localization is symmetric monoidal and exact. Since S^n is finite, it commutes with $C^*(S^n; -)$ and with the trivial square-zero extension. Consequently, $C^*(S^n; R)$ is $\mathbb{E}_{n+1}$ -trivial over R . The contrapositive of [5, Theorem 4.12, equivalently Propositions 4.20 and 4.22] gives

$$H\mathbb{F}_p \otimes R = 0, \quad KU/p \otimes R = 0. \quad (5.1)$$

The spectra KU/p and $K(1, p)$ have the same Bousfield class, as used in [5, proof of Lemma 4.13]. Indeed, p -local periodic complex K -theory splits as the sum of $p - 1$ suspensions of its periodic Adams summand, whose reduction modulo p is a model of $K(1, p)$. Thus (5.1) implies $K(1, p) \otimes R = 0$.

The spherical triviality of R and the two vanishings $H\mathbb{F}_p \otimes R = K(1, p) \otimes R = 0$ now satisfy the three implications of Proposition 5.4. They give $R/p = A_{(p)}/p = 0$.

Rationality of A . The global Moore spectrum $\mathbb{S}_{\mathrm{global}}/p$ is already p -local. To see this, let q be an integer coprime to p . Multiplication by q induces an isomorphism on its integral homology. The cofiber of this map is bounded below and has zero integral homology, so the stable Hurewicz theorem makes it zero. Smashing with A shows that A/p is also p -local. Hence

$$A/p \simeq (A/p)_{(p)} \simeq A_{(p)}/p = 0.$$

As this holds at every prime, multiplication by every nonzero integer is an equivalence on A . Each homotopy group of A is therefore a rational vector space. The rationalization formula

$$\pi_*(A \otimes H\mathbb{Q}) = \pi_* A \otimes \mathbb{Q}$$

now shows that the unit-induced map $A \rightarrow A \otimes H\mathbb{Q}$ is an equivalence.

The rational implication. Conversely, suppose that A is rational. Its unit factors through $H\mathbb{Q}$ as an $\mathbb{E}_{n+2}$ -map. By [5, Lemma 4.10], with $m = n + 1$, the cochains $C^*(S^n; A)$ are equivalent to the trivial square-zero extension $A \oplus \Sigma^{-n} A$ as an augmented $\mathbb{E}_{n+1}$ - A -algebra. $\square$

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